



COURSE CODE/TITLE:	CPE 212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	09/01/2026	DATE SUBMITTED:	09/01/2026
NAME:	Melquisedec D Iquin	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 6 Targeting Specific Nodes

1. PROCEDURE OUTPUT

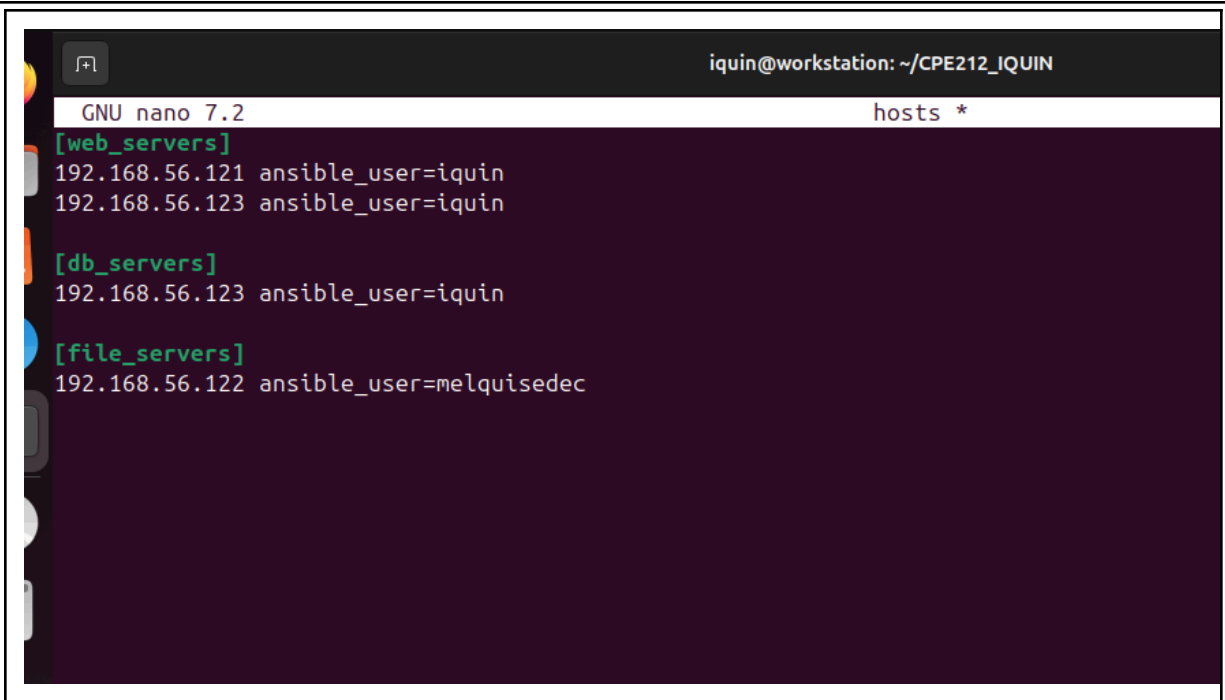
What procedures did you perform?

```
GNU nano 7.2 site.yml
--
- hosts: all
  become: true
  tasks:
    - name: install apache and php for Ubuntu servers
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache and php for CentOS server
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"
```

```
iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 site.yml *
---
- hosts: all
  become: true
  pre_tasks:
    - name: install updates (CentOS)
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"
    - name: Install updates (Ubuntu)
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:
    - name: install apache and php for Ubuntu servers
      apt:
        name:
          - apache2
          - libapache2-mod-php
File Name to Write: site.yml
^G Help          M-D DOS Format  M-A Append
^C Cancel        M-M Mac Format  M-P Prepend
Right Ctrl
```

A screenshot of a terminal window with a dark background. The title bar at the top shows a window icon on the left and the text 'iquin@workstation: ~/CPE212_IQUIN' on the right. The terminal content shows the GNU nano 7.2 editor interface. The first line is 'hosts *'. Below it, there are three sections: '[web_servers]' with two lines '192.168.56.121 ansible_user=iquin' and '192.168.56.123 ansible_user=iquin'; '[db_servers]' with one line '192.168.56.123 ansible_user=iquin'; and '[file_servers]' with one line '192.168.56.122 ansible_user=melquisedec'.

```
iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 hosts *
[web_servers]
192.168.56.121 ansible_user=iquin
192.168.56.123 ansible_user=iquin

[db_servers]
192.168.56.123 ansible_user=iquin

[file_servers]
192.168.56.122 ansible_user=melquisedec
```

GNU nano 7.2

sit

```
---  
  
- hosts: all  
  become: true  
  pre_tasks:  
  
    - name: install mariadb package (CentOS)  
      yum:  
        name: mariadb-server  
        state: latest  
        when: ansible_distribution == "CentOS"  
  
    - name: "Mariadb- Restarting/Enabling"  
      service:  
        name: mariadb  
        state: restarted  
        enabled: true  
  
- name: install updates (Ubuntu)  
  apt:  
    upgrade: dist  
    update_cache: yes  
    when: ansible_distribution == "Ubuntu"  
  
- hosts: web_servers  
  become: true  
  tasks:
```



```
iquin@workstation: ~/CPE212_IQUIN

TASK [install mariadb package (CentOS)] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
ok: [192.168.56.122]

TASK [Mariadb- Restarting/Enabling] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
changed: [192.168.56.122]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.122]
ok: [192.168.56.123]
changed: [192.168.56.121]

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]
ok: [192.168.56.123]

TASK [install apache and php for Ubuntu servers] *****
ok: [192.168.56.121]
ok: [192.168.56.123]

PLAY RECAP *****
192.168.56.121      : ok=4    changed=1    unreachable=0    failed=0    skipped=2    resc
192.168.56.122      : ok=3    changed=1    unreachable=0    failed=0    skipped=1    resc
192.168.56.123      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    resc
```

```
GNU nano 7.2                                site.yml *  
  
- hosts: web_servers  
  become: true  
  tasks:  
    - name: install apache and php for Ubuntu servers  
      apt:  
        name:  
          - apache2  
          - libapache2-mod-php  
        state: latest  
        when: ansible_distribution == "Ubuntu"  
  
- hosts: file_servers  
  become: true  
  tasks:  
    - name: install samba package  
      package:  
        name: samba  
        state: latest  
  
Save modified buffer?   
Y Yes  
N No      ^C Cancel
```

```

iquin@workstation: ~/CPE212_IQUIN
ok: [192.168.56.121]
ok: [192.168.56.123]

TASK [install apache and php for Ubuntu servers] *****
ok: [192.168.56.121]
ok: [192.168.56.123]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.122]

TASK [install samba package] *****
changed: [192.168.56.122]

PLAY RECAP *****
192.168.56.121      : ok=4    changed=0    unreachable=0    failed=0    s
skipped=2    rescued=0    ignored=0
192.168.56.122      : ok=5    changed=2    unreachable=0    failed=0    s
skipped=1    rescued=0    ignored=0
192.168.56.123      : ok=4    changed=0    unreachable=0    failed=0    s
skipped=2    rescued=0    ignored=0

iquin@workstation:~/CPE212_IQUIN$

```

TASK 2

```
GNU nano 7.2                                site.yml
---
- hosts: all
  become: true
  pre_tasks:

  - name: install updates (CentOS)
    tags: always
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    tags: always
    apt:
      upgrade: dist
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - hosts: web_servers
    become: true
    tasks:
      - name: install apache and php for Ubuntu servers
        tags: apache, apache2, ubuntu
        [ Read 63 lines ]

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 site.yml
when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  tags: apache, centos, httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  - name: install mariadb package (CentOS)
    tags: centos, db, mariadb
    dnf:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "Ubuntu"
```

```
iquin@workstation: ~/CPE212_IQUIN
iquin@workstation:~/CPE212_IQUIN$ sudo nano site.yml
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts site.yml --ask-pass --ask-become-pass
SSH password:
BECOME password[defaults to SSH password]:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.122]
ok: [192.168.56.121]

TASK [install updates (CentOS)] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
ok: [192.168.56.122]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.122]
ok: [192.168.56.121]
ok: [192.168.56.123]

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]
ok: [192.168.56.123]
```



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```
iquin@workstation: ~/CPE212_IQUIN

TASK [Gathering Facts] *****
ok: [192.168.56.121]
ok: [192.168.56.123]

TASK [install apache and php for Ubuntu servers] *****
ok: [192.168.56.121]
ok: [192.168.56.123]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]

TASK [install mariadb package (CentOS)] *****
skipping: [192.168.56.123]

TASK [Mariadb- Restarting/Enabling] *****
skipping: [192.168.56.123]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.122]
```

```
iquin@workstation: ~/CPE212_IQUIN
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts --tags centos --ask-become-pass
site.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.121]
ok: [192.168.56.122]

TASK [install updates (CentOS)] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
ok: [192.168.56.122]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.122]
ok: [192.168.56.123]
ok: [192.168.56.121]

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.121]
```

2.1

```
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook --list-tags site.yml
[WARNING]: Could not match supplied host pattern, ignoring: web_servers
[WARNING]: Could not match supplied host pattern, ignoring: db_servers
[WARNING]: Could not match supplied host pattern, ignoring: file_servers

playbook: site.yml

  play #1 (all): all    TAGS: []
    TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
    TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers    TAGS: []
    TASK TAGS: [centos, db, mariadb]

  play #4 (file_servers): file_servers    TAGS: []
    TASK TAGS: []
```

2.2

```
iquin@workstation: ~/CPE212_IQUIN
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts --tags centos --ask-become-pass site.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.121]
ok: [192.168.56.122]

TASK [install updates (CentOS)] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
ok: [192.168.56.122]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.122]
ok: [192.168.56.123]
ok: [192.168.56.121]

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.121]
```

2.3

```
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts --tags db --ask-become-pass site.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]
ok: [192.168.56.122]
ok: [192.168.56.123]

TASK [install updates (CentOS)] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
ok: [192.168.56.122]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.122]
ok: [192.168.56.121]
ok: [192.168.56.123]

PLAY [web_servers] *****
```

2.4

```
iquin@workstation: ~/CPE212_IQUIN
run operations as this user (default=root)
-b, --become run operations with become (does not imply password prompting)
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts --tags apache --ask-become-pass
site.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.122]
ok: [192.168.56.121]

TASK [install updates (CentOS)] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
ok: [192.168.56.122]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.122]
ok: [192.168.56.121]
ok: [192.168.56.123]

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]

TASK [install apache and php for Ubuntu servers] *****
```

2.5

```
iquin@workstation: ~/CPE212_IQUIN
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts --tags "apache.db" --ask-become
-pass site.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.121]
ok: [192.168.56.122]

TASK [install updates (CentOS)] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
ok: [192.168.56.122]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.122]
ok: [192.168.56.123]
ok: [192.168.56.121]

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
```

PART 3

```
iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 site.yml *
tasks:
- name: install apache and php for Ubuntu servers
  tags: apache,apache2,ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  tags: apache,centos,httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags: apache,centos,httpd
  service:
    name: httpd
    state: started
  when: ansible_distribution == "CentOS"

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```
iquin@workstation: ~/CPE212_IQUIN
rescued=0    ignored=0

iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts site.yml --ask-pass --ask-become-pass
SSH password:
BECOME password[defaults to SSH password]:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.122]
ok: [192.168.56.121]

TASK [install updates (CentOS)] *****
skipping: [192.168.56.121]
skipping: [192.168.56.123]
ok: [192.168.56.122]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.122]
ok: [192.168.56.121]
ok: [192.168.56.123]

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]
```

```
iquin@workstation: ~/CPE212_IQUIN
ok: [192.168.56.121]

TASK [install apache and php for Ubuntu servers] *****
ok: [192.168.56.121]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.121]

TASK [start httpd (CentOS)] *****
skipping: [192.168.56.121]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.122]

TASK [install mariadb package (CentOS)] *****
ok: [192.168.56.122]

TASK [Mariadb- Restarting/Enabling] *****
changed: [192.168.56.122]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]

TASK [install samba package] *****
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

```
iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 roles/webserver/tasks/main.yml
--
- name: install apache and php for Ubuntu servers
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  service:
    name: httpd
    state: started
  when: ansible_distribution == "CentOS"

[ Read 23 lines ]
```

```
iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 roles/database/tasks/main.yml *
--
- name: install mariadb package (CentOS)
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: "Mariadb- Restarting/Enabling"
  service:
    name: mariadb
    state: restarted
    status: true
  when: ansible_distribution == "CentOS"
```

```
iquin@workstation: ~/CPE212_IQUIN
[file_servers]
192.168.56.123 ansible_user=iquin ansible_ssh_pass=Qwerty@123
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts site.yml --tags web --ask-become-pass
BECOME password:

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]

TASK [webserver : install apache and php for Ubuntu servers] *****
ok: [192.168.56.121]

TASK [webserver : install apache and php for CentOS servers] *****
skipping: [192.168.56.121]

TASK [webserver : start httpd (CentOS)] *****
skipping: [192.168.56.121]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.122]

PLAY RECAP *****
192.168.56.121      : ok=2    changed=0    unreachable=0    failed=0    skipped=2
   rescued=0    ignored=0
192.168.56.122      : ok=1    changed=0    unreachable=0    failed=0    skipped=0
```

```
iquin@workstation: ~/CPE212_IQUIN
iquin@workstation:~/CPE212_IQUIN$ nano roles/database/tasks/main.yml
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts site.yml --tags db --ask-become-pass
BECOME password:

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.122]

TASK [database : install mariadb package (CentOS)] *****
ok: [192.168.56.122]

TASK [database : Mariadb- Restarting/Enabling] *****
changed: [192.168.56.122]

PLAY RECAP *****
192.168.56.121      : ok=1    changed=0    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0
192.168.56.122      : ok=3    changed=1    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

iquin@workstation:~/CPE212_IQUIN$
```

```

iquin@workstation: ~/CPE212_IQUIN
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts site.yml --ask-become-pass
BECOME password:

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.121]

TASK [webserver : install apache and php for Ubuntu servers] *****
ok: [192.168.56.121]

TASK [webserver : install apache and php for CentOS servers] *****
skipping: [192.168.56.121]

TASK [webserver : start httpd (CentOS)] *****
skipping: [192.168.56.121]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.122]

TASK [database : install mariadb package (CentOS)] *****
ok: [192.168.56.122]

TASK [database : Mariadb- Restarting/Enabling] *****
changed: [192.168.56.122]

PLAY RECAP *****

```

```
iquin@workstation: ~/CPE212_IQUIN
ok: [192.168.56.121]

TASK [webserver : install apache and php for Ubuntu servers] *****
ok: [192.168.56.121]

TASK [webserver : install apache and php for CentOS servers] *****
skipping: [192.168.56.121]

TASK [webserver : start httpd (CentOS)] *****
skipping: [192.168.56.121]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.122]

TASK [database : install mariadb package (CentOS)] *****
ok: [192.168.56.122]

TASK [database : Mariadb- Restarting/Enabling] *****
changed: [192.168.56.122]

PLAY RECAP *****
192.168.56.121      : ok=2    changed=0    unreachable=0    failed=0    skipped=2
rescued=0    ignored=0
192.168.56.122      : ok=3    changed=1    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0
iquin@workstation:~/CPE212_IQUIN$
```

Reflections:

Answer the following:

1. What is the importance of putting our remote servers into groups?
 - So putting our remote servers into groups matters because it lets me target specific sets of machines instead of running every task on every server. In my inventory file, I organized my servers into web_servers, db_servers, and file_servers, so my playbook only installs apache on the web servers and mariadb on the database server.
2. What is the importance of tags in playbooks?
 - The tags are important because they let me run only specific parts of my playbook instead of executing the whole thing every time. I used tags: web and tags: db on my roles, so I could run just the webserver setup or just the database setup separately with --tags web or --tags db as instructed.
3. Why do think some services need to be managed automatically in playbooks?
 - I think some services need to be managed automatically in playbooks because doing it manually across multiple servers is slow and easy to mess up, I experienced this myself with typos along with moving the file wrong to the root directory which is very very frustrating and a lot of typos and missed underscores and the indentation.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

gemini

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

- *Purpose of this use is to check my grammar*

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.